TEXORA · LECTURE SHEET

Recycled Fiber

Textile waste is a fibre resource — if it is sorted, opened back to fibre and blended honestly. This sheet covers mechanical recycling of pre- and post-consumer waste, what tearing does to fibre quality, blending with virgin fibre, and the chemical-recycling route for polyester.

YM-4103 · SPECIAL YARN MANUFACTURING · TOPIC 10 · LEVEL 4 TERM 1 · YARN MANUFACTURING SPECIALIZATION

1. Where recycled fibre comes from

Recycled fibre is fibre recovered from textile material that has already been made once, and returned to a new spinning or nonwoven process. Sources divide into two classes, and the class decides the quality that is realistically available:

- **Pre-consumer (post-industrial) waste** — spinning mill waste, weaving and knitting off-cuts, garment cutting waste and selvedges. It has never been worn or washed, its fibre type and colour are usually known, and it is clean. This is the best recycling feedstock and the main source of spinnable recycled fibre.
- **Post-consumer waste** — used garments and household textiles collected after service. It is mixed in fibre, colour and condition, may carry buttons, zips, prints, finishes and contamination, and needs far more sorting and cleaning. Much of it can only be downcycled into coarse yarns, nonwovens, insulation or filling unless sorting is excellent.

A third distinction matters for claims: fibre recovered and respun into a lower-value product is *downcycling*; returning it to an equivalent product needs the best feedstock and careful blending. Honest grading of the input — not optimism — sets the product.

2. Mechanical recycling — the tearing line

Mechanical recycling recovers fibre by physically pulling fabric apart, without dissolving or melting it. The standard line runs as follows:

1. **Sorting.** Waste is sorted by fibre type, colour and construction. Sorting by colour avoids re-dyeing; sorting by fibre avoids mixing fibres that spin and dye differently. Trims — buttons, zips, rivets,

elastic — are removed, by hand, by machine, or both; a single missed metal part damages the tearing machinery.

2. **Cutting.** The sorted fabric is guillotine- or rotary-cut into small pieces of even size, so the tearing machine is fed uniformly and no large piece can wrap or choke the rollers.
3. **Tearing / shredding (opening).** The pieces pass through a series of pinned or saw-tooth tearing rollers (a "rag tearing" line, historically called shoddy/mungo production for wool). Each stage pulls the woven or knitted structure apart progressively — yarns first separate, then yarns open into fibres. Lubricant or water spray may be added to reduce fibre breakage and dust.
4. **Cleaning and dedusting.** Air separation and beating remove dust, short fragments and remaining non-fibrous material.
5. **Blending.** The recovered fibre is blended with virgin fibre (Section 3) and, where needed, with fibre of other colours to hit the target shade.
6. **Carding and spinning.** The blend is carded — often on a woollen-type or adapted cotton card — and spun, most successfully by open-end (rotor) spinning for apparel-range counts (Section 4).

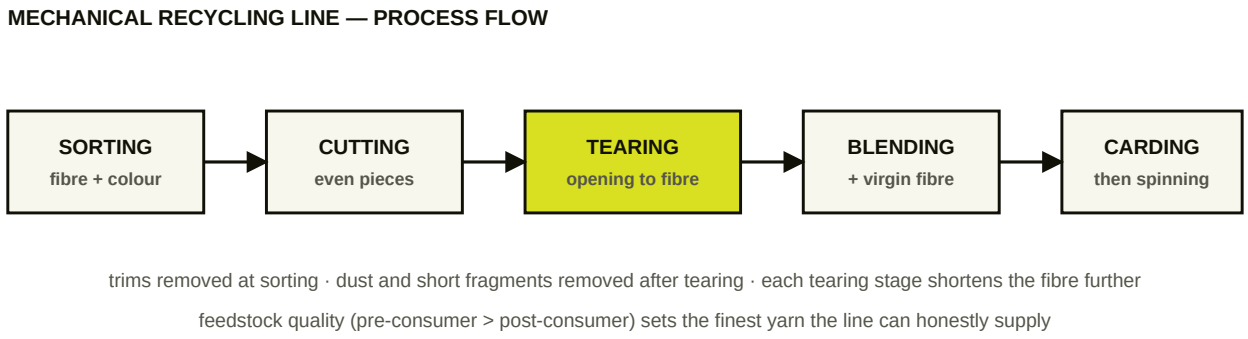


Figure 1 — Mechanical recycling line. Sorted textile waste is cut, torn back to fibre, blended with virgin fibre and carded for spinning.

3. Quality consequences and blending with virgin fibre

Tearing is violent work, and the fibre pays for it. The consequences must be designed for, not discovered afterwards:

- **Fibre shortening.** Every tearing stage breaks fibres. Mean fibre length falls substantially — as typical values, cotton recovered from fabric commonly comes back with a mean length well below that of new cotton (for example of the order of 12–18 mm against roughly 25–30 mm for virgin cotton; treat these as typical ranges, varying with feedstock and line severity). Shorter fibre means fewer fibres can be held by twist over any cross-section, so yarn strength and spinnable fineness fall.

- **Strength loss.** Yarns spun from 100% recovered staple fibre are typically markedly weaker and more irregular than virgin-fibre yarns of the same count — again as a typical statement, a strength reduction of the order of 20–40% is commonly seen, depending on blend and feedstock. Neps and remaining yarn fragments raise irregularity further.
- **Colour and contamination limits.** Mixed-colour input gives mixed, muted shades; fine sorting by colour is labour-intensive. Residual finishes and fibre mixtures limit dyeing levelness.

The standard answer is **blending with virgin fibre**. Longer, stronger virgin fibres form a load-bearing framework through the yarn, lifting strength, spinnability and evenness, while the recycled fraction supplies bulk, colour and the environmental benefit. Blend ratio is a specification decision: coarse open-end yarns and denims can carry high recycled proportions; finer or more stressed yarns carry less. The blend must also be intimate — recycled and virgin fibre are blended before or at carding, never merely twisted together, or the yarn varies along its length.

> **WORKED EXAMPLE — BLEND CALCULATION**
>
> A 30 tex open-end yarn is to be spun from a blend of 60% recycled cotton and 40% virgin cotton.
> Per 100 kg of yarn: recycled cotton = 0.60 × 100 = **60 kg**; virgin cotton = 0.40 × 100 = **40 kg**.
> For a 500 kg production lot at the same ratio: recycled = 0.60 × 500 = **300 kg**; virgin = 0.40 × 500 = **200 kg**.
> In practice the feeder weighs slightly more than these figures to allow for carding and spinning waste — the ratio, not the exact input mass, is what the blend calculation fixes.

4. Chemical recycling — outline

Chemical recycling recovers the polymer, not the fibre, so it can in principle restore full quality. For **PET polyester** the route is depolymerisation: collected PET (from bottles, and increasingly from polyester textiles) is broken down by glycolysis, methanolysis or hydrolysis into its monomers or intermediates, purified — purification is the point: dyes and contaminants leave with the residue — and then re-polymerised into new PET, which is melt-spun into fibre indistinguishable in quality from virgin-fibre polyester. For **cotton and cellulosics**, dissolution routes dissolve the cellulose and regenerate it as a manufactured cellulosic fibre. Chemical routes cost more energy and capital than mechanical tearing and are still limited in textile-to-textile scale, but they are the only routes that escape the fibre-shortening limit of mechanical recycling.

5. Applications

- **Open-end (rotor) yarns.** Rotor spinning tolerates shorter fibre better than ring spinning, so recycled blends are spun mainly as OE yarns in coarse-to-medium counts — denim weft, casual knitwear,

towels, fleece and workwear fabrics.

- **Nonwovens.** Recovered fibre that is too short or mixed to spin is needled or bonded into insulation, padding, automotive felts, geotextiles, mops and wipes — large, honest downcycling outlets that absorb most post-consumer material.
- **Filling and low-grade products.** The shortest residues go to flock, filling and shoddy-type products rather than to landfill.
- **Premium recycled yarns.** From clean, colour-sorted pre-consumer waste, blended with virgin fibre, branded recycled-cotton yarns reach normal apparel use — the blend ratio and feedstock are declared, which is what makes the claim credible.

The selection rule closes the sheet: match the recycled stream to the product it can honestly serve. Forcing short, mixed post-consumer fibre into a fine, stressed yarn produces a weak yarn and discredits recycling; placing it in OE yarns and nonwovens produces sound products at scale.

Glossary

Recycled fibre — fibre recovered from previously manufactured textile material. **Pre-consumer waste** — clean mill and cutting waste that never reached a user. **Post-consumer waste** — textiles recovered after use. **Tearing / shredding** — mechanical opening of fabric back to fibre on pinned rollers. **Shoddy / mungo** — traditional names for fibre torn from woollen rags. **Downcycling** — recycling into a lower-value product. **Depolymerisation** — chemical breakdown of a polymer to monomers for purification and re-polymerisation. **Open-end (rotor) spinning** — spinning system tolerant of shorter fibres, widely used for recycled blends.

Key points

• Pre-consumer waste is clean and known — the best feedstock; post-consumer waste is mixed and needs heavy sorting and cleaning.
• Mechanical line: sorting (trims out) → cutting → tearing → cleaning → blending → carding → spinning.
• Tearing shortens fibre and yarn strength falls (typical values only — confirm by test); blending with virgin fibre restores spinnability and strength.
• Chemical recycling (e.g. PET depolymerisation and re-polymerisation) restores polymer quality but costs more and is still limited in scale.
• Main outlets: open-end yarns, nonwovens, filling — matched honestly to the quality of the recovered stream.

Practice

1. A mill plans 800 kg of OE yarn at 45% recycled cotton / 55% virgin cotton. Calculate the mass of each component in the yarn. State one reason the feeder input would be weighed slightly higher.
2. Explain, in process order, why a missed zip in sorted garments is dangerous, and at which line stage it must be removed.
3. A buyer demands a fine, high-strength ring-spun shirting yarn from 100% post-consumer mixed garments. Write a short technical refusal: name the two fibre-level consequences of tearing that make the demand unrealistic, and propose a realistic product instead.
4. Distinguish mechanical and chemical recycling of polyester in three respects: what is recovered, what happens to fibre length/quality, and what happens to dyes and contaminants.

Answer key

1. Recycled = 0.45 × 800 = **360 kg**. Virgin = 0.55 × 800 = **440 kg**. Input is weighed higher to allow for processing waste (carding and spinning losses), so that the finished yarn mass and ratio are met.
2. Zips contain metal/plastic teeth and sliders. If they reach the tearing rollers they damage the pinned clothing, spark and contaminate the fibre with fragments. They must be removed at the **sorting** stage, before cutting and tearing — the first stage of the line.
3. Tearing (1) shortens the fibres severely, so too few long fibres remain for a fine ring-spun yarn to hold together, and (2) the mixed, abraded fibre gives a weak, irregular yarn — strength falls typically by the order of 20–40% even in blends. A realistic product: a coarser open-end yarn blended with virgin cotton, or a nonwoven product, where the recovered fibre's length and strength are sufficient.
4. Mechanical: the *fibre* is recovered by tearing; the fibre is shortened and its quality falls; dyes stay in the fibre (hence colour sorting). Chemical: the *polymer* is recovered by depolymerisation; it is purified and re-polymerised, so new fibre of full length and virgin-like quality can be spun; dyes and contaminants are removed during purification.
